Can Machines Read Manga?

Detecting and Recognizing Stylized Text in Japanese Manga

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Motivations

- To a human (like me), manga is a rich reading experience enhanced by dialogue and imagery
- To a machine (unlike me), manga is a dense representation of textual and visual information
- Sound effects are both textual and visual, and contribute to the reading experience

- Also, I like reading and translating manga :)

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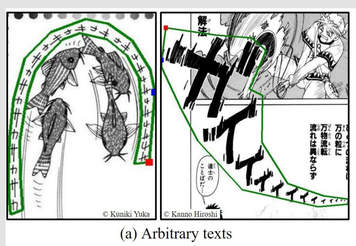
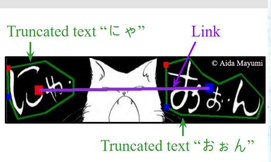
English and Japanese Language Releases of *My Hero Academia*

Introduction

- Sound effects in both Japanese manga and Western comics are heavily stylized with color, kerning, font, and hand drawn characters
- Reading them is akin to two tasks; detection and recognition
  - Detect where text is expected
  - Recognize the characters and/or words detected
- Some key differences between stylized text in manga and standard text detection include:
  - Arbitrary texts
  - Truncated texts
  - Rotated and/or vertical texts



(a) Arbitrary texts



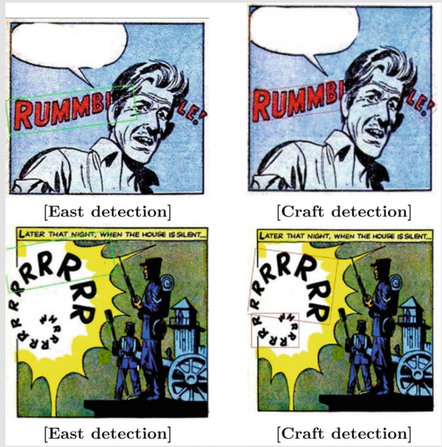
(b) Truncated texts and link between them

Figure demonstrating arbitrary text placement (a) and truncated texts (b), Baek et al. [2]

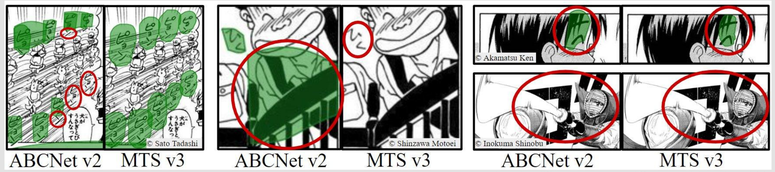
Image 14-4 of the KABOOM Dataset outlined by Louis et al. [1]

Experiments

- Purpose-built datasets, like the comic onomatopoeia dataset (COO) and KABOOM onomatopoeia dataset (KABOOM)
- Different algorithms for text detection and recognition:
  - ABCNet v2 (COO)
  - MTS v3 (COO)
  - EAST (KABOOM)
  - CRAFT (KABOOM)
- Different techniques to enhance results:
  - Rotation
  - SAR decoding
  - Hard ROI Masking

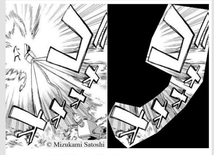


Text detection differences between different models Louis et al. [1]

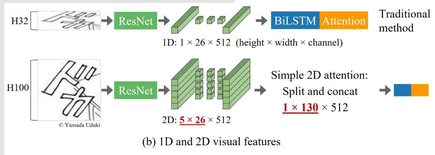


Text detection differences between different models Baek et al. [2]

- Results varied based on the model, techniques applied, and dataset training
- CRAFT > EAST (Character level detection versus word level detection)
  - Both algorithms saw improvement when trained on KABOOM
- MTS v3 > ABCNet v2 (Segmentation versus Regression)
  - Despite ABCNet v2 performing better on standard benchmarks
- One of the key issues of text recognition is recognizing vertical texts
- Text recognition generally occurs left to right, so some tricks explored by Baek et al. [2]:
  - Rotation +3.5%
  - SAR decoding -3.1%
  - Both Rot. & SAR +9.1%
  - Add Hard RoI +21.6%
  - Add 2D Attention +34.7%



(a) Hard RoI masking



(b) 1D and 2D visual features

Figure describing RoI Masking (a) and 2D Attention (b), Baek et al. [2]

Conclusion

- Stylized sound effects in manga continue to present a challenging computer vision task
- Pushing progress through purpose-built datasets, nuanced models, and combined recognition techniques
- Despite these advances, some techniques actively remove context from the manga to aid in text detection and recognition
- Large language models, vision language models , and large multimodal models may be able to navigate the abstracted context to improve detection and recognition [3]
- "Comprehension" is a key part to reading sound effects in manga to me, it may be the case for a computer too

Sound effects found in *Harukaze Mound*

References

[1] J. B. Louis, J.-C. Burie, and A. Revel, "Can Deep Learning Approaches Detect Complex Text? Case of Onomatopoeia in Comics Albums," in *Pattern Recognition, Computer Vision, and Image Processing. ICPR 2022 International Workshops and Challenges*, J.-J. Rousseau and B. Kapralos, Eds., Cham: Springer Nature Switzerland, 2023, pp. 48–60. doi: 10.1007/978-3-031-37742-6_4.

[2] J. Baek, Y. Matsui, and K. Aizawa, "COO: Comic Onomatopoeia Dataset for Recognizing Arbitrary or Truncated Texts," in *Computer Vision – ECCV 2022*, S. Avidan, G. Brostow, M. Cissé, G. M. Farinella, and T. Hassner, Eds., Cham: Springer Nature Switzerland, 2022, pp. 267–283. doi: 10.1007/978-3-031-19815-1_16.

[3] Y. Chen, Y. Ren, and S. Süsstrunk, "Zooming into Comics: Region-Aware RL Improves Fine-Grained Comic Understanding in Vision-Language Models," Nov. 09, 2025, arXiv: arXiv:2511.06490. doi: 10.48550/arXiv.2511.06490.

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